

Fun by Design: Visual Form, Realism, and the Hedonic Appeal of AI-Powered Virtual Assistants

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ABSTRACT

Despite ongoing debates about the benefits and challenges of AI in retail, advances in AI continue to drive the evolution of virtual assistants (VAs). Initially limited to simple, formless chatbots, VAs have progressively incorporated visual representations and, more recently, hyper-realistic, human-like digital personas enabled by AI. While these innovations have significantly enhanced the form and realism of VAs, their true impact remains underexplored, with mixed evidence on how such visual improvements influence consumer perception and, ultimately, VA service quality. Complicating this issue is a persistent theoretical bias in the existing literature, which predominantly depends on technology-, social-, and cognitive-centered frameworks. Although technology-based theories were useful in early VA adoption, their explanatory power declines as consumer familiarity grows. Moreover, the online retail context—where VAs primarily operate—limits the relevance of traditional social and cognitive theories. Unlike goal-driven, socially interactive in-store shopping, online retail is often solitary, self-paced, and hedonic. Online consumers browse not only to fulfill functional or cognitive needs but also to seek leisure, escape from daily routines (liberation), and indulge in immersive pleasure (hedonic engagement), treating shopping as a source of fun and stress relief. Given these dynamics, VA research must move beyond traditional frameworks to better capture the hedonic nature of modern online retail. To address these gaps, we examine how VA form and realism influence consumer evaluations of VA services, focusing on the mediating roles of liberation, hedonic engagement, and fun, as outlined in consumer fun theory. Using VA stimuli generated by AI tools, we conducted three experimental studies. Study 1 found that simply adding a visual form to VA (compared to no form) significantly improved consumer evaluations, mediated by liberation, hedonic engagement, and fun. Study 2 showed that increasing VA form realism (high vs. low) further amplified service evaluations through the same mediation pathways. Study 3 confirmed these effects and revealed that recreation-oriented shoppers experienced stronger mediation effects than task-oriented shoppers. Together, these findings offer fresh theoretical and practical insights for enhancing VA design and deployment in online retail environments.

1. Introduction

Generative AI has revolutionized the retail landscape, with *virtual assistant (VA) services* becoming one of its most transformative

applications. Powered by advanced AI, VAs have redefined how online retailers interact with and serve their customers (Flavián, Akdim, & Casalo, 2023). Designed to simulate the functions of human shopping assistants, VAs perform a range of tasks, including answering customer

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inquiries, providing personalized product recommendations, assisting with post-purchase services such as returns, and offering real-time support to enhance the overall shopping experience (Belk, Belanche, & Flavián, 2023; Rese, Ganster, & Baier, 2020). Despite these clear advantages, concerns persist regarding potential drawbacks associated with this technological shift. Critics caution about risks such as the displacement of human customer service representatives (Belanche, Belk, Casalo, & Flavián, 2024; Dwivedi et al., 2021) and the possible erosion of the personal touch in customer service (Flavián, Tusyadiah, & Orús, 2024). Nonetheless, many proponents argue that the benefits of VA services—including cost efficiency, improved operational effectiveness, and greater consumer convenience—outweigh these concerns, viewing VAs as a positive and enduring development in retail technology (Dwivedi et al., 2023).

Building on this optimistic outlook, VA interfaces have evolved rapidly in tandem with advancements in AI, becoming increasingly capable of replicating not only human behavior but also human appearance. Early VA designs were relatively rudimentary, consisting primarily of text- or voice-based chatbots without any visual components (see Table A in the Web Appendix). During this initial stage, design efforts focused on improving human-like behavior—such as tone, responsiveness, and conversational flow—to create more engaging interactions (Bawack, Wamba, & Carillo, 2021; Li, Li, Chen, & Chang, 2024). As AI matured, however, attention turned toward visual embodiment. Developers began crafting VAs with human-like appearances to improve realism and relatability (Keeling, McGoldrick, & Beatty, 2010). This shift led to the rise of visually embodied VAs, ranging from simple avatar-style figures to hyper-realistic digital humans, with clear applications in retail. For example, IKEA uses a basic animated avatar-style VA named *Anna* to assist customers, while Firework—a livestreaming commerce platform—features a hyper-realistic VA to support online customer service (see Table B in the Web Appendix). These cases illustrate how the evolution of VA appearance—from formless chatbots to basic avatars and ultimately to highly realistic embodiments—is being strategically leveraged by retailers, with an intention to strengthen service quality and enhance overall effectiveness.

Despite these significant commercial advancements, academic research has not kept pace. Our review of VA-related studies—detailed further in Section 2.1—reveals a widening gap between the accelerating evolution of VA applications in industry and the relatively static focus and scope of academic research (Figure C in the Web Appendix). This disconnect is particularly evident across three critical areas: contextual bias, an anthropomorphism-related imbalance, and theoretical bias. First, the literature demonstrates a pronounced *contextual bias*. Although VA design and visual embodiment have advanced considerably over the past five years, most academic studies remain focused on “formless VAs”—especially text-based chatbots—while only a small minority examine “visually embodied VAs” or “hyper-realistic VAs” (Table D in the Web Appendix). This disproportionate focus fails to reflect the growing prevalence of visually anthropomorphized VAs in commercial practice. As a result, much existing research is misaligned with real-world developments and offers limited guidance for practical implementation, design refinement, or consumer engagement in contemporary retail contexts.

Second, our review revealed a persistent *imbalance in the representation and prioritization of anthropomorphism* in the VA literature (Table E in the Web Appendix). Anthropomorphism—defined here as the attribution of human-like qualities to nonhuman agents, including how realistically a VA’s cognition, behavior, and appearance mirror those of humans—is widely regarded as a critical factor shaping consumer responses. Although anthropomorphism spans multiple dimensions, academic research in VA has disproportionately focused on two in particular: “cognitive realism” (i.e., the extent to which a VA exhibits human-like thinking, emotions, and decision-making) and “behavioral realism” (i.e., the extent to which its actions and behaviors resemble

those of humans). These are often treated as the dominant indicators of VA performance. In contrast, “form realism” (i.e., the extent to which its physical appearance resembles that of a human) has received comparatively little attention, despite its growing relevance in real-world retail applications (Pentina, Xie, Hancock, & Bailey, 2023). This imbalance not only mirrors the earlier contextual biases but is further complicated by inconsistent findings in the limited research on form realism. While some studies associate higher form realism with more favorable consumer responses (Song, Zhu, Xing, & Du, 2024), others suggest it may evoke discomfort or unease (Pizzi, Scarpi, & Pantano, 2021). These mixed results make it difficult to derive clear design implications for retail practitioners aiming to enhance service quality through VA embodiment. Consequently, further empirical investigation is needed to determine how varying levels of form realism influence consumer perceptions and overall service outcomes.

Third, our review uncovered a notable *theoretical bias* in existing VA research (Table F in the Web Appendix). Although VAs have become increasingly integrated into everyday consumer life, most studies continue to rely heavily on traditional “technology-oriented frameworks,” such as the technology acceptance model, as their primary theoretical lens. These are followed by “social-oriented theories,” like social response theory, and “cognitive-oriented theories,” such as the elaboration likelihood model. While these frameworks have provided valuable early insights, their dominance limits the theoretical diversity needed to fully capture the evolving and multifaceted nature of consumer-VA interactions in contemporary digital commerce. Notably, many of these frameworks were developed for contexts that differ substantially from the realities of online shopping. Unlike the socially embedded and goal-oriented nature of in-store retail, online shopping is often solitary, self-paced, and hedonically driven. Consumers often browse alone not only to fulfill functional or cognitive needs but also to escape daily routines (liberation), and derive personal enjoyment (hedonic engagement)—treating shopping as a form of entertainment (fun) or leisure (Kim, Kim, & Park, 2010; Koch, Frommeyer, & Schewe, 2020b). Given these dynamics, the theoretical frameworks guiding VA research must transcend traditional technology-, social-, and cognitive-centered approaches to fully incorporate the hedonic, emotional, and experiential dimensions that define today’s online retail environment.

To address these contextual, anthropomorphic, and theoretical limitations, we center our investigation on visually embodied VAs, with a particular focus on form realism—an underexplored yet crucial dimension of VA anthropomorphism that warrants deeper examination. We adopt consumer fun theory (Oh & Pham, 2022) to frame our approach within a more experience-centered and affective perspective. The theory posits that modern consumers increasingly value experiences that evoke “fun,” a state arising from fulfilling two core psychological needs: a sense of liberation (i.e., a temporary escape from psychological constraints) and hedonic engagement (i.e., an active participation in pleasurable activities). We argue that consumers will seek this same sense of fun—rooted in liberation and hedonic engagement—when interacting with VAs. Moreover, acknowledging that these interactions take place in an online shopping context—where consumer motivations vary, encompassing task-oriented goals (i.e., shopping with the intent to purchase) and recreation-oriented goals (i.e., shopping for enjoyment and leisure)—we investigate how these distinct motivations may moderate consumer responses to VAs. To empirically examine these dynamics, our research is structured into three studies:

- Study 1 investigates how the form of a VA—its absence versus visual presence—affects consumers’ service evaluations. It examines both the main effect (VA form → service evaluation) and two serial mediation pathways: (1) VA form → perceived liberation → perceived fun → service evaluation, and (2) VA form → perceived hedonic engagement → perceived fun → service evaluation. By revealing how the visual presence of a VA enhances perceived service

quality through a chain of experiential responses rooted in fun, this study addresses important contextual and theoretical gaps in existing VA research.

- Study 2 builds on these findings by investigating the role of VA visual anthropomorphism—operationalized as low versus high form realism—in shaping consumers' service evaluations. It examines both the main effect (VA form realism → service evaluation) and two serial mediation pathways: (1) VA form realism → perceived liberation → perceived fun → service evaluation, and (2) VA form realism → perceived hedonic engagement → perceived fun → service evaluation. By shifting the focus of the VA literature from its disproportionate emphasis on cognitive and behavioral realism to form realism, this study highlights the importance of visual sophistication in enhancing consumer perceptions of service quality through fun-driven experiential responses.
- Study 3 replicates the main effect and serial mediation effects of VA form realism observed in Study 2 and further explores the moderating role of consumers' shopping motivations (task-oriented vs. recreation-oriented). By examining how different goal orientations shape responses to VA design, this study provides deeper insight into how VAs can be strategically tailored to align with varying shopping contexts and consumer profiles.

2. Theoretical background and hypotheses development

2.1. Key literature in virtual assistant research

While the introduction briefly highlighted key literature gaps, we now offer a more detailed review. To thoroughly understand current knowledge and limitations regarding VAs in online retail, we conducted a comprehensive review of high-quality studies published between 2020 and 2025. This timeframe was chosen to capture the most recent advancements in the field. We focused exclusively on studies published in SSCI-indexed journals and ranked "A" or above on the ABDC journal list, ensuring that our insights are grounded in the most credible findings. Figure C in the Web Appendix illustrates details of the review procedure. This rigorous review allowed us to uncover three critical research gaps, which served as the foundation for the design of our study.

First, the review revealed a strong *contextual bias* (Table D in the Web Appendix). Despite significant advancements in VA visual embodiment driven by AI within the online retail sector over the past five years, most VA research (87 %, $n = 34$) focused on "formless VAs," primarily chatbots (Adam & Benlian, 2024; Cai, Li, & Law, 2022) and voice assistants (Hsu & Lee, 2023; Kautish, Purohit, Filieri, & Dwivedi, 2023). In contrast, only 8 % of the studies ($n = 3$) have examined "visually embodied VAs," such as animated avatar-based agents (Ben Saad & Choura, 2024). An even smaller fraction (5 %, $n = 2$) has investigated "VAs with form realism," exploring differences in anthropomorphic appearance ranging from low to high (Song et al., 2024; Yang, Peng, Wang, Zhao, & Wang, 2025). This disproportionate focus underscores a significant disconnect between academic research and real-world practice. The scarcity of scholarly attention to these embodied formats limits our understanding of their actual and potential impacts on consumer engagement and retailer performance. Bridging this gap necessitates systematic inquiry into the comparative advantages of visually embodied VAs over formless ones, as well as the incremental benefits conferred by enhanced humanlike appearance in retail contexts.

Second, the review revealed a notable *imbalance in how anthropomorphism is conceptualized and operationalized* within VA research (Table E in the Web Appendix). Anthropomorphism emerged as a pivotal explanatory construct in over half of the reviewed studies (20 out of 39), commonly categorized into three dimensions: "cognitive realism," "behavioral realism," and "form realism." Among these, *cognitive realism* has received the most extensive scholarly attention over the past five years ($n = 14$). This body of research often contrasts different VA capabilities—such as thinking VA versus feeling VA—and examines how

these distinct functions shape user experiences. Evidence consistently shows that feeling VAs tend to elicit stronger user engagement compared to their purely cognitive counterparts (Huang & Rust, 2018, 2021; Rust & Huang, 2021; Schepers, Belanche, Casaló, & Flavián, 2022). Cheng, Bao, Zarifis, Gong, and Mou (2021) further demonstrated that a VA's capacity to express emotion, particularly empathy, significantly increases users' trust, ultimately leading to greater reliance on these agents. *Behavioral realism* has also received steady attention ($n = 9$), with research focusing on facets such as linguistic anthropomorphism (e.g. machine-like vs. human-like; Yang et al., 2025) and variations in language style (e.g., formal vs. informal; Li & Wang, 2023), aiming to clarify which behavioral cues most effectively enhance VA performance.

In contrast, *form realism*—while immediately noticeable and impactful due to its visual attributes—has received the least attention in academic research ($n = 3$; Table E in the Web Appendix). Not only are studies surprisingly few, but the existing findings on its effects are inconclusive, leading to ambiguity and ongoing debate within the literature. For instance, one study suggests that increased human-likeness in a VA's appearance may trigger user discomfort, leading to less favorable service evaluations (Song et al., 2024). However, another study finds that higher levels of form realism can enhance user experience by amplifying social presence, which in turn fosters greater engagement and more positive evaluations (Yang et al., 2025). This inconsistency signals a clear need for further research—not only to clarify these mixed findings but also to help bridge the contextual gap where VAs with visible form and high form realism remain underexplored.

Third, the review uncovered a notable *theoretical bias* in existing VA research (Table F in the Web Appendix). The most commonly adopted frameworks were "technology-oriented theories" ($n = 11$), such as the unified theory of acceptance and use of technology (Fu, Mouakket, & Sun, 2024; Silva, De Cicco, Vlačić, & Elmashhara, 2023) and the technology acceptance model (Chen, Le, & Florence, 2021; Nayak & Nair, 2025), which primarily view VAs as functional tools—emphasizing usability and efficiency. This was followed by "social-oriented theories" ($n = 8$), including social response theory (Li & Wang, 2023; Song et al., 2024) and the computers as social actors theory (Cheng, Fan, Shao, Jia, & Zhang, 2024; Song, Xing, Duan, Cohen, & Mou, 2022), which focus on the interpersonal dynamics between consumers and VAs. "Cognitive-oriented theories" ($n = 6$), such as the elaboration likelihood model (Weiler, Matt, & Hess, 2022; Yang et al., 2025), were also employed to examine how consumers process information from VAs and how this affects persuasion and decision-making. Aligned with these theoretical frameworks, frequently examined *mediators* included "performance expectancy" under technology-oriented theories (Fu et al., 2024; Vimal-kumar, Sharma, Singh, & Dwivedi, 2021), "parasocial interaction" under social-oriented theories (Adam & Benlian, 2024; Jiang, Yang, & Zheng, 2023), and "cognitive trust" or "perceived information quality" under cognitive-oriented theories (Bawack et al., 2021; Vimalkumar et al., 2021).

While these theoretical perspectives and respective mediators offer valuable insights into specific aspects of consumer-VA interactions—such as functionality, social engagement, and cognitive evaluation—they fall short of fully addressing the distinctive characteristics of the online retail environment for three main reasons. First, growing consumer familiarity and comfort with VA technology have diminished the explanatory power of traditional technology-focused theories in understanding consumer interactions with VAs. Second, unlike physical shopping, which is often interactive, online shopping tends to be a solitary, self-directed activity. Consumers usually browse and shop alone via mobile devices or computers (Van Nguyen, McClelland, & Thuan, 2024). This solitary nature is intensified by the convenience and easy access of online platforms, allowing consumers to shop anytime and anywhere without the need for social interaction. As a result, social-oriented theories that emphasize social presence and interpersonal interaction may have limited applicability in explaining

consumer–VA interactions in online retail. Third, unlike traditional retail—where shopping is usually goal-driven and information-focused—online shopping is often motivated by hedonic and experiential desires (Santo & Marques, 2022). Consumers visit online stores not only to purchase but also to seek relief from daily pressures (liberation) and enjoy immersive, pleasurable experiences (hedonic engagement), treating online shopping as a source of fun and leisure rather than a purely task-focused activity (Kim et al., 2010; Koch, Frommeyer, & Schewe, 2020a). Therefore, cognitive-oriented theories that emphasize rational information processing and decision-making may not fully capture the emotional drivers of consumer behavior in online contexts, including interactions with VAs during the process. Collectively, these three limitations underscore the need to broaden the theoretical lens in VA research beyond the traditional technology-, social-, and cognitive-oriented frameworks. In particular, it is essential to integrate approaches that better account for consumers' hedonic motivations in online retail. One promising framework is consumer fun theory (Oh & Pham, 2022).

2.2. Consumer fun theory

In today's fast-paced and often stressful world, the pursuit of “fun” has become a central theme in modern consumer life (Oh & Pham, 2018). With the constant juggling of work and daily tasks, people often experience significant stress. To counterbalance these pressures, they actively seek opportunities for “having fun,” which broadly refers to activities that bring subjective feelings of pleasure and enjoyment (Oh & Pham, 2022). This desire for fun is often expressed through behaviors, such as indulging in gourmet food, engaging in leisure activities, or shopping, each offering an escape from daily stressors and providing moments of hedonistic pleasure.

To gain deeper insight into why some consumption experiences are perceived as more fun than others, Oh and Pham (2018) introduced consumer fun theory. In this framework, fun is defined as “an experience of liberating engagement” (Oh & Pham, 2022, p. 47), emphasizing that the enjoyment derived from an experience is rooted in two key psychological factors: hedonic engagement and liberation. The first factor, *hedonic engagement*, involves immersive participation aimed purely at enjoyment, such as losing oneself in browsing an online store or exploring products to buy (Lee & Youn, 2024). When consumers fully engage in these moments, they perceive the experience as fun. However, hedonic engagement alone is not enough. For an experience to be truly enjoyable, it must also involve *liberation* (Oh & Pham, 2018). This second factor, liberation, refers to the temporary release from the psychological constraints of daily life (Brigham, 2005). This sense of escape, often enabled through consumption, offers relief and freedom, thereby heightening the feeling of fun. Consumer fun theory posits that it is the synergistic combination of hedonic engagement and liberation that produces the most gratifying and enjoyable consumption experiences.

Building on this theoretical foundation, our study investigates whether two psychological pathways—liberation and hedonic engagement—jointly shape consumers' perceptions of fun and serve as key mediators in their evaluation of VAs. We specifically hypothesize that the form and visual realism of a VA influence consumers' service evaluations through two distinct serial mediation processes: (1) VA form → perceived liberation → perceived fun → VA service evaluation, and (2) VA form → perceived hedonic engagement → perceived fun → VA service evaluation. By examining how visual embodiment and realism contribute to the creation of fun experiences, and, more importantly, how these experiences enhance consumers' evaluations of VA services, our study offers new insights into the role of visually embodied VAs in facilitating pleasurable, immersive interactions that align with the hedonic nature of online shopping. In doing so, we not only address key contextual and theoretical gaps in existing VA research but also reconceptualize VAs as providers of experiential and hedonic value in online retail, rather than as merely functional tools.

2.3. Hypotheses development

Before presenting the serial mediation hypotheses, we first propose the main effect of a VA's form on consumers' service evaluations, establishing the foundation for examining the more complex mediated effects.

2.3.1. How VAs' “form” influences consumer evaluation

The positive influence of VAs' *visual representations*—specifically, the absence versus presence of a visual form—on consumer perceptions is well documented in the existing literature. Qiu and Benbasat (2009) demonstrated that consumers are more willing to interact with a VA that features a male, animated-style avatar (i.e., presence of visual form condition) than one that uses only text without any facial representation (i.e., absence of visual form condition). Similarly, Go and Sundar (2019) observed that incorporating a female human-like form into a VA interface, rather than merely displaying a dialogue bubble without a defined form, resulted in more favorable consumer attitudes toward the website offering the VA service. These findings highlight the significant role that a VA's visual form plays in shaping consumer evaluations of the service. Building upon these insights, we propose:

H1a. A VA with a form (compared to one without) will lead to more positive consumer evaluations of the VA service.

2.3.2. How VAs' “form realism” influences consumer evaluation

Beyond merely having a visual form, we propose that the *degree of realism in that form* also plays a crucial role in shaping consumers' perceptions of VAs. This idea is well supported by avatar marketing theory (Miao, Kozlenkova, Wang, Xie, & Palmatier, 2022) and anthropomorphism theory (Epley, Waytz, & Cacioppo, 2007). According to avatar marketing theory, VAs with a more human-like appearance (i.e., high-form realism condition) are more effective at delivering superior online services compared to those with less human-like features (i.e., mid- or low-form realism conditions) or nonhuman representations (Miao et al., 2022). For example, Miao et al. (2022) found that VAs with 3D human-like images are seen as more effective at improving the customer experience and service quality compared to those with 2D, cartoonish designs. Similarly, anthropomorphism theory explains that people tend to attribute human-like qualities to nonhuman entities, so the more human-like a VA appears, the more positive consumers perceive it, which enhances engagement with the service (Epley et al., 2007).

Empirical studies further support these theoretical frameworks. For example, Song et al. (2024) found that a highly anthropomorphic VA with a male profile led to a greater intention to engage with the VA compared to a less human-like, robotic counterpart. Likewise, Letheren, Jetten, Roberts, and Donovan (2021) discovered that consumers exhibit a strong preference for and more favorable evaluations of VAs that closely resemble human beings, as opposed to those with a more mechanical, robotic appearance. Research on virtual influencers further underscores the positive impact of form realism on consumer evaluations and marketing effectiveness. Specifically, higher levels of form realism in virtual influencers have been associated with increased consumer engagement (Ju, Kim, & Im, 2024; Park & Sung, 2023) and more favorable evaluations of the products they endorse (Dabiran, Farivar, Wang, & Grant, 2024; Kim, Ki, Lee, & Kim, 2024). Moreover, additional studies show that virtual influencers with high (vs. low) form realism can enhance brand-related outcomes, such as generating more positive brand attitudes (Ozdemir, Karabulut, & Messenger, 2025), heightened consumer engagement (Barari, Ross, & Azad Moghddam, 2025), and increased purchase intentions (Zourrig, Park, & Becheur, 2025). Collectively, these theoretical perspectives and empirical findings from adjacent domains underscore the favorable influence of form realism in shaping consumer evaluations. Extending this to our research context, we propose:

H1b. A VA with high form realism (compared to one with low form realism) will lead to more positive consumer evaluations of the VA service.

2.3.3. How VAs' form enhances consumer service evaluation through "liberation" followed by "fun"

As previously discussed, the sense of *liberation* refers to the temporary release from external stressors, allowing individuals to experience relief and freedom (Oh & Pham, 2022). Existing research suggests that the presence of a VA with a form—particularly one with a high level of form realism—can significantly enhance this sense of liberation among consumers. Feste, Peters, and Schirmer (2023) found that consumers interacting with an online retail service featuring an avatar form (i.e., a VA with a visual form), as opposed to no avatar form (i.e., a VA with no form), reported greater stress relief and felt more liberated during their online shopping experience. Similarly, Ki, Chong, Aw, Lam, and Wong (2025) showed that consumers engaging with highly realistic avatars (i.e., high form realism), compared to less realistic ones (i.e., low form realism), experienced a more effective mental escape from real-life stressors, thereby strengthening their perceived sense of liberation. Expanding on these findings, we hypothesize that VAs with a form (compared to those without) and with more human-like forms (compared to less realistic ones) will foster stronger feelings of liberation in consumers.

Furthermore, prior research suggests that feelings of liberation can enhance perceptions of *fun*. For example, Oh and Pham (2020) provided empirical evidence demonstrating that activating a sense of liberation can increase consumers' enjoyment. Kim and Baek (2025) further supported this relationship by showing that a sense of liberation positively impacts perceived humor—one manifestation of fun—in social media contexts. Building on these findings, we propose that a similar effect will occur in the VA context. Therefore, we hypothesize that the more consumers feel liberated during their interaction with a VA, the more fun they will find the experience.

Finally, we posit that the more enjoyable (i.e., fun) the interaction with a VA, the more positively consumers will evaluate the VA service and the more likely they will be to reengage with it in the future. This hypothesis is supported by previous research demonstrating that the fun derived from a consumption experience can significantly enhance consumers' evaluations of both the experience and the associated service, resulting in higher satisfaction and more favorable appraisals of the service (Nguyen, Nguyen, & Barrett, 2007; Slåtten, Mehmetoglu, Svensson, & Sværi, 2009). Based on these insights, we hypothesize that the influence of a VA's form—encompassing both its presence and level of realism—on consumers' service evaluations is sequentially mediated by feelings of liberation and fun, which subsequently foster more favorable outcomes. Accordingly, we propose the following:

H2a. A VA with a form (compared to one without) will evoke stronger feelings of liberation and fun, subsequently resulting in more favorable evaluations of the VA service.

H2b. A VA with high form realism (as opposed to low form realism) will evoke stronger feelings of liberation and fun, subsequently resulting in more favorable evaluations of the VA service.

2.3.4. How VAs' form enhance consumer service evaluation through "hedonic engagement" followed by "fun"

In addition to proposing liberation and fun as key serial mediators in the relationship between VA form and consumer service evaluation, we introduce another important serial mediation pathway through *hedonic engagement* and *fun*. As mentioned earlier, hedonic engagement refers to the active involvement and immersion in pleasurable experiences (Oh & Pham, 2020). Research suggests that virtual agents with a visual form evoke higher levels of hedonic immersion than those without. For example, Jin (2010) found that participants who interacted with a virtual agent featuring a form in an online test perceived the test experience

as more informative and immersive compared to those who completed the test without a virtual agent. Beyond the mere presence of a VA form, its level of realism also plays a significant role. Barney, Hancock, Jones, Kazandjian, and Collier (2022) demonstrated that, in the context of retail apps, virtual agents with anthropomorphic features—such as a face or other human-like elements—evoke greater consumer immersion than those without such characteristics. Building on these findings, we propose that VAs with a form (compared to those without) and with more human-like visual forms (compared to less realistic ones) will foster stronger feelings of hedonic engagement in consumers.

Extant research also underscores the significant relationship between hedonic engagement and the perception of *fun*. Papagiannidis, Pantano, See To, Dennis, and Bourlakis (2017) suggested consumers are more likely to perceive an experience as enjoyable when they are deeply immersed in it. Similarly, numerous studies have demonstrated that flow—a state similar to hedonic engagement where individuals become highly focused and absorbed in an activity—induces feelings of enjoyment (Murphy, Chertoff, Guerrero, & Moffitt, 2014; Sweetser & Wyeth, 2005). Even in gaming contexts, researchers assert that while a player's perception of fun is subjective, it is closely tied to the psychological flow concept (Velikovskiy, 2014). Given the parallels between flow and hedonic engagement, and prior research linking flow to fun, we extend this to our context and hypothesize that the more hedonic engagement consumers experience during interaction with a VA, the more fun they will perceive.

Finally, extensive research across various contexts demonstrates that enhancing consumers' sense of fun during an experience leads to more positive evaluations toward the associated brand or service. Scarpi, Pizzi, and Visentin (2014) demonstrated that the more fun consumers have while shopping, the greater their loyalty and positive attitude toward the retailer. Similarly, Karl, Harland, Peluchette, and Rodie (2010) found that when employees experience greater fun in their work—particularly in health care settings—it results in higher satisfaction not only for the employees themselves but also for the patients they engage with. Building on these insights, we propose that greater fun experienced during interactions with a VA will lead to more favorable consumer evaluations of the VA service. Based on these insights, we hypothesize that the effect of a VA's form—both its presence and realism—on consumers' service evaluations is sequentially mediated by feelings of hedonic engagement and fun, which subsequently lead to more favorable evaluations. Thus, we propose the following serial mediation hypotheses:

H3a. A VA with a form (compared to one without) will evoke stronger feelings of hedonic engagement and fun, subsequently resulting in more favorable evaluations of the VA service.

H3b. A VA with high form realism (as opposed to low form realism) will evoke stronger feelings of hedonic engagement and fun, subsequently resulting in more favorable evaluations of the VA service.

2.3.5. How consumers' "online shopping motivation" influences their responses to virtual assistant form realism

Just as the visual form of a VA—its presence and realism—can influence consumers' perceptions and behaviors in the online retail context, we further propose that consumers' situational shopping motivations also play a role in shaping these outcomes. Shopping motivation—defined as a consumer's overall attitude or mindset toward the shopping process (Moon, Farooq, & Kiran, 2017)—has been widely recognized in retail services literature as a significant moderating factor. Kaltcheva and Weitz (2006) categorized these shopping motivations into two primary types: *task-oriented* and *recreation-oriented*. Task-oriented shoppers approach shopping based on the immediate situational need, treating it as a goal-directed activity aimed at acquiring essential products, services, or information. In this context, the primary focus is on fulfilling a specific need rather than enjoying the shopping process, with any enjoyment being secondary or minimal (Büttner,

Florack, & Göritz, 2015). On the other hand, recreation-oriented shoppers view shopping as an enjoyable, immersive activity (Kaltcheva & Weitz, 2006). For these consumers, the shopping experience itself is pleasurable, driven by the desire for entertainment or leisure. They derive enjoyment from activities like browsing products, regardless of whether a purchase is made. Thus, for recreation-oriented consumers, shopping becomes a situational experience focused on seeking enjoyment rather than fulfilling a specific goal.

Even in online retail, consumers exhibit distinct motivations when visiting retailers. Some consumers approach with a clear intent to make a purchase (Childers, Carr, Peck, & Carson, 2001), while others browse purely for enjoyment without any immediate intent to buy (Scarpi et al., 2014). Recreation-oriented consumers focus less on achieving specific outcomes and more on deriving pleasure from the shopping process itself. They enjoy exploring new arrivals, seeking inspiration, and discovering unique or hidden products simply for fun, treating online shopping as a leisure activity fueled by curiosity and the joy of discovery. In contrast, task-oriented shoppers visit online stores with the goal of fulfilling a specific need or purchase (Wenzel & Benkenstein, 2018). As demonstrated across this body of research, these underlying shopping motivations significantly shape consumers' emotional responses and behavioral outcomes in retail environments. Accordingly, we expect that the effects of VA form realism on service evaluation—mediated through liberation, hedonic engagement, and fun—will be more pronounced for recreation-oriented consumers than for their task-oriented counterparts. Thus, we hypothesize:

H4. The serial mediation effects of VA form realism on service evaluation via liberation, hedonic engagement, and fun will be stronger for recreation-oriented than for task-oriented consumers.

Fig. 1 provides a visual representation of our conceptual model.

3. Overview of studies

To evaluate the proposed hypotheses, we conducted three online experiments. Study 1 used a one-way between-subjects design to examine the impact of a VA's visual form (absence vs. presence) on consumers' evaluations of VA services. This study tested both the main effect of VA form on service evaluation (H1a) and two serial mediation pathways: whether the effect of VA form on service evaluation is

mediated by liberation and fun (H2a), and by hedonic engagement and fun (H3a). Study 2 also employed a one-way between-subjects design but manipulated the realism of the VA's visual form (low vs. high). Building on Study 1's findings, it assessed whether greater realism in VA appearance enhances consumer evaluations, focusing on both its main effect (H1b) and indirect effects through the same serial mediation processes: VA form realism → liberation → fun → service evaluation (H2b), and VA form realism → hedonic engagement → fun → service evaluation (H3b). Study 3 adopted a 2×2 between-subjects design, manipulating both VA form realism (low vs. high) and online shopping motivation (task-oriented vs. recreation-oriented). The goals of this study were to replicate the findings of Study 2 (H1b, H2b, and H3b), and to examine whether shopping motivation moderates the two serial mediation effects (H4). Table 1 summarizes each study's objectives and design.

4. Study 1

4.1. Stimuli and procedures

4.1.1. Stimuli development using generative AI: form (absence vs. presence)

To manipulate the VA form into two distinct conditions—(1) the “absence” condition and (2) the “presence” condition—we developed two video stimuli, each based on a standardized text scenario and different VA representations. In the absence condition, the VA was represented solely by voice, emulating a traditional, nonvisual interface. By contrast, in the presence condition, we used generative AI to produce a video featuring a modern VA interface with an anime-style digital avatar that greets and interacts with consumers.

We opted for video stimuli over static images or mock apps for two key reasons: (1) unlike static images, videos allow participants not only to observe the VA's visual appearance but also experience its interactive service in a more natural and fluid way, and (2) unlike mock apps, video stimuli allow us to control every aspect of the presentation—such as timing, pacing, and interaction flow—minimizing the potential influence of extraneous variables on the participants. This approach enabled us to isolate and examine the true impact of VA form on consumer evaluations. Additionally, to address concerns from those who might believe that videos are less immersive than mock apps, we included an open-ended question at the end of the survey. As shown in Table H in the

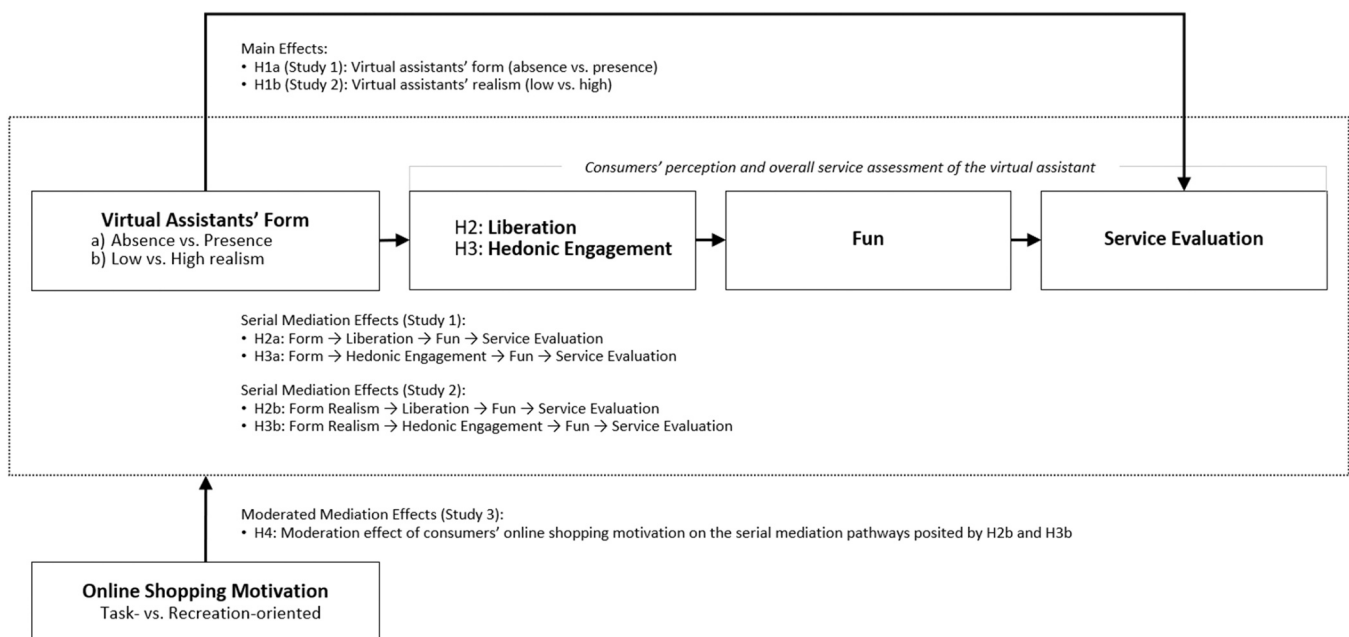


Fig. 1. Conceptual Model.

Table 1
Summary of the Experimental Design for Study 1, 2, and 3.

	Study 1	Study 2	Study 3
Purpose	- H1a: To test the main effect of a VA's form on consumers' service evaluation. - H2a: To test the serial mediation effect of liberation and fun on the relationship between the VA form and consumers' service evaluation. - H3a: To test the serial mediation effect of hedonic engagement and fun on the relationship between the VA's form and consumers' service evaluation.	- H1b: To test the main effect of a VA's form on realism on consumers' service evaluation. - H2b: To test the serial mediation effect of liberation and fun on the relationship between the VA form realism and consumers' service evaluation. - H3b: To test the serial mediation effect of hedonic engagement and fun on the relationship between the VA's form realism and consumers' service evaluation.	- H1b, H2b, and H3b: To test the robustness of the main and serial mediation effects between VA form realism and consumers' service evaluation. - H4: To examine the moderation effect of consumers' online shopping motivation on the serial mediation effects stated in H2b and H3b.
Independent variable	VA visual form: - Absence - Presence	VA form realism: - Low - High	VA form realism: - Low - High
Dependent variable	Service evaluation	Service evaluation	Service evaluation
Mediators	- Liberation - Hedonic engagement - Fun	- Liberation - Hedonic engagement - Fun	- Liberation - Hedonic engagement - Fun
Moderator	N/A	N/A	- Online shopping motivation: - Task-oriented motivation - Recreation-oriented motivation
Experimental design	- A one-way between-subjects design (VA visual form: absence vs. presence)	- A one-way between-subjects design (VA form realism: low vs. high)	- A 2 (VA form realism: low vs. high) × 2 (Consumers' shopping motivation: task vs. recreation) between-subjects design
Stimuli development using generative AI	Video stimuli designed to manipulate VA form and VA form realism: - Images featured in the video: - Images of an online fashion store and the high form realism VA were generated using Bing's AI Image Creator. - The low-form realism VA used in Study 2 (which corresponded to the 'visual form presence' condition in Study 1) was created from the high form realism VA image using the Voila AI Artist app. - Textual scenario used in the video: - Textual scenarios, including greetings, information about new arrivals, and	Video stimuli designed to manipulate VA form realism: - The video stimuli from Study 2 (low-form realism and high-form realism conditions) were used. Textual scenario designed to manipulate consumers' online shopping motivation:	

Table 1 (continued)

	Study 1	Study 2	Study 3
	promotional details, were developed according to the VA's service protocol. - Narration and animation: - VAs for all conditions (absence, presence, low form realism, and high form realism) were narrated with AI-generated speech using HeyGen. - VAs with a visual interface (presence, low form realism, and high form realism conditions) were further animated using HeyGen.		- Two textual scenarios (task- and recreation-oriented conditions), depicting consumers' different shopping motivations for visiting an online fashion store, were developed.
Conditions and sample size	- Absence (n = 90) - Presence (n = 89)	- Low FR (n = 150) - High FR (n = 150)	- Low FR × Task (n = 85) - Low FR × Recreation (n = 84) - High FR × Task (n = 84) - High FR × Recreation (n = 85)

Note. Virtual assistant (VA); Form realism (FR).

Web Appendix, the responses from participants across both stimuli conditions demonstrate that they fully engaged with the content and understood the intended service interaction.

The details of the video stimuli development are outlined further here. We began by designing a VA *scenario* for the video, following a standardized service protocol (see Table I in the Web Appendix), which included customer greetings, information on new arrivals, and promotional details. This uniform scenario was applied across both conditions to ensure consistency. We next employed generative AI to produce two sets of *images*: (1) a background image of the fashion retail store, and (2) a visual representation of the VA. To create the background image of a fashion store, we used Bing's AI Image Creator with the prompt: "Create a high-resolution image of a retail store for a non-luxury fashion brand, such as Zara or H&M, named Livia Style." The AI-generated image was refined until it accurately reflected a typical fashion retail environment.

We then used Bing's AI Image Creator again to generate a visual representation of the VA for the presence condition. Specific keywords, such as "female assistant in a non-luxury fashion retail store, depicted from the upper body, American, friendly demeanor, introducing herself to customers," were provided to guide the AI in creating an image that mirrored real-world VA examples. After obtaining the AI-generated image (which was later used in the *high form realism* condition in Study 2), we further transformed it into a more cartoonish version using the Voila AI Artist app, drawing inspiration from IKEA's animated VA, Anna (Table B in the Web Appendix).

Next, we employed HeyGen, an AI-integrated video creation tool, to combine the textual scenario, images, and AI-generated audio narration, resulting in the final two distinct *video* stimuli. In the absence condition, the VA's visual form was omitted, retaining only the AI-generated voice and the background store image. In contrast, the presence condition included an animated VA with synchronized lip movements and facial expressions that matched the AI-generated speech (see Table I). No manipulation check was necessary in Study 1, as the presence or absence of the VA's visual form in the video stimuli was inherently clear and did not require further validation.

4.1.2. Main test: survey design and data collection

We created an online survey for the main test, structured into four sections: (1) consent and introduction, (2) stimulus presentation, (3) main variable questions, and (4) demographic information. In the first section, participants were asked to provide consent and were presented with definitions and examples of VAs in the retail domain to ensure

clarity regarding the research context. After reviewing this information, participants were randomly assigned to one of two (absence vs. presence) video conditions. Before being exposed to the randomly assigned stimuli, participants were provided with the following instruction: “You will encounter a video where an AI-powered virtual retail assistant guides you through an online fashion retail store. Please watch attentively and provide your honest opinions and perceptions in response to the questions following the video.” In the second section, participants were required to watch the video for a minimum of 50 seconds to ensure full engagement. To further verify participants’ attention to the critical stimuli, an attention check question was included: “What is the gender of the VA?” Only participants who answered this question correctly were allowed to proceed to the next section. In the third section, participants provided their assessments of the interaction with the assigned VA, evaluating key factors such as perceived liberation (Oh & Pham, 2022), hedonic engagement (Oh & Pham, 2022), and fun (Oh & Pham, 2022; Tasci & Ko, 2016) following their exposure to the video stimulus. Additionally, they rated their overall service evaluation of the VA (Ki, Cho, & Lee, 2020; Kim & Hur, 2023). All items were measured on a 7-point Likert scale, ranging from “strongly disagree” (1) to “strongly agree” (7). To ensure participants’ continuous attention throughout the questionnaire, we included an additional attention check asking them to select “somewhat agree” at the center of the scale. Participants who did not select this option were additionally screened out and any responses they provided were excluded from further analysis. The specific measurement items employed in this study are detailed in Table 2. In the final section of the survey, participants were asked to provide their demographic information.

The survey was conducted through Dynata, a professional marketing survey company, targeting US residents aged 18 and older. After

excluding invalid responses, 179 remained: 90 in the absence condition and 89 in the presence condition. Power analysis confirmed the sample size exceeded the minimum ($n = 127$) needed for 13 predictors at $\alpha = .05$, 80 % power, and a medium effect size of .15 (Cohen, 1988). Most respondents were female (50 %), aged above 55 (33 %), Caucasian (76 %), married (48 %), held a bachelor’s degree (36 %), were employed full-time (54 %), and had an annual household income in the 75,000 – 149,999 USD range (36 %).

4.2. Results

4.2.1. Common method bias test results

To address potential common method bias, we employed the marker variable technique (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003) using the lavaan package in R. Specifically, we included a theoretically unrelated marker variable—attitude toward the color blue (Miller & Simmering, 2023)—measured using seven items on a 7-point Likert scale ranging from “strongly disagree” (1) to “strongly agree” (7). This marker variable was incorporated into the confirmatory factor analysis (CFA) as a separate latent construct with its own indicators. Following the procedures outlined by Williams, Hartman, and Cavazotte (2010), we compared the model fits of four CFA models incorporating the orthogonal marker variable: (1) the baseline model, with all loadings from the marker variable to the substantive indicators constrained to zero; (2) the Method-C model, with all such loadings constrained to be equal; (3) the Method-U model, allowing the loadings to vary freely; and (4) the Method-R model, identical to Method-U but with inter-construct correlations fixed to the values from the baseline model. Chi-square difference tests revealed a significant difference between the baseline and Method-C models ($\Delta\chi^2 = 15.01$, $p < .001$), indicating the presence

Table 2
Construct Reliability and Convergent Validity Assessment Results.

Scale items	Study 1				Study 2				Study 3			
	Loading	α	AVE	CR	Loading	α	AVE	CR	Loading	α	AVE	CR
<i>Form realism</i>						.952	.878	.956		.946	.864	.950
This virtual assistant looks real and natural.					.983				.978			
This virtual assistant looks realistic.					.986				.979			
This virtual assistant in its entirety seemed to be authentic for this occasion.					.835				.823			
<i>Liberation</i>		.930	.869	.930		.914	.842	.914		.899	.817	.899
Through engaging with this virtual assistant, I felt delight that outweighed my worries.	.931				.931				.905			
The experience with this virtual assistant made me feel liberated.	.931				.904				.903			
<i>Hedonic engagement</i>		.929	.814	.929		.941	.843	.941		.924	.808	.927
Interacting with this virtual assistant provided me with a sense of engagement.	.928				.906				.903			
I was absorbed in the experience of engaging with this virtual assistant.	.869				.933				.939			
I truly felt present and “in the moment” during my engagement with this virtual assistant.	.909				.915				.853			
<i>Fun</i>		.935	.831	.936		.941	.848	.934		.932	.828	.935
Engaging with this virtual assistant was an immensely fun experience.	.892				.932				.932			
Engaging with this virtual assistant provided me with a lot of fun moments in this experience.	.939				.949				.944			
Engaging with this virtual assistant gave me pleasure.	.902				.880				.852			
<i>Service evaluation</i>		.981	.912	.981		.983	.922	.983		.985	.930	.985
I am interested in utilizing this virtual assistant service again if there is an opportunity.	.959				.964				.964			
I intend to utilize this virtual assistant service in the future if there is an opportunity.	.966				.974				.969			
I would recommend this virtual assistant service to my friends.	.919				.937				.944			
I plan to use this virtual assistant service in the future if there is an opportunity.	.971				.959				.972			
I expect to use this virtual assistant service in the future if there is an opportunity.	.959				.965				.972			

Note. Cronbach’s alpha (α); average variance extracted (AVE); composite reliability (CR).

of some method variance. However, no significant differences emerged between Method-C and Method-U ($\Delta\chi^2 = 10.89$, $p = .54$), or between Method-U and Method-R ($\Delta\chi^2 = 0.29$, $p = 1.00$). These results indicate that while method variance exists, it is non-congeneric and does not meaningfully bias the estimates of the substantive constructs. Therefore, common method bias was not a major concern in Study 1.

4.2.2. Measurement test results

With common method bias ruled out, we proceeded to evaluate the measurement model. The CFA results demonstrated a satisfactory model fit: $\chi^2 = 104.22$, $df = 59$, $\chi^2/df = 1.77$, CFI = .99, TLI = .98, NFI = .97, and RMSEA = .07 (Gheshlagh et al., 2018). As shown in Table 2, all factor loadings exceeded the .71 threshold, indicating strong construct reliability (Tabachnick, Fidell, & Ullman, 2013). Cronbach's alpha (α), composite reliability (CR), and average variance extracted (AVE) values surpassed the recommended cut-offs of .70, .60, and .50, respectively, supporting convergent validity (Cheung, Cooper-Thomas, Lau, & Wang, 2023; Sarmiento & Costa, 2017). Additionally, discriminant validity was confirmed using the Fornell-Larcker criterion (Table 3; Cheung et al., 2023).

4.2.3. Hypotheses test results

After validating the measurement model, we tested H1a, H2a, and H3a. One-way ANOVA results revealed a significant main effect of VA form (absence = 0, presence = 1) on VA service evaluation ($F[1, 177] = 8.03$, $p < .01$), providing support for H1a. To further validate this main effect, we conducted a mediation analysis using Hayes (2015) PROCESS macro (Model 80, bootstrapping with $n = 5,000$ resamples) as well. As illustrated in Table 4, the main effect was significant (total effect: $\beta = .73$, 95 % CI = [.22, 1.24]). In addition to the significant main effect, we confirmed two significant serial mediation effects: one through liberation and fun (indirect effect: $\beta = .19$, 95 % CI = [.06, .36]) and another through hedonic engagement and fun (indirect effect: $\beta = .10$, 95 % CI = [.04, .20]). In the form presence (vs. absence) condition, consumers reported significantly higher levels of liberation ($\beta = .54$, $p < .001$) and hedonic engagement ($\beta = .42$, $p < .01$). These increases in liberation ($\beta = .58$, $p < .001$) and hedonic engagement ($\beta = .38$, $p < .001$) subsequently led to higher perceptions of fun, which in turn resulted in more favorable service evaluations ($\beta = .34$, $p < .001$). Together, these results support H2a and H3a.

4.3. Discussion

The findings yielded by Study 1 offer compelling preliminary evidence highlighting the importance of a VA's form (as opposed to a formless VA) in enhancing consumers' evaluations of the VA service. Specifically, the visual form not only had a positive main effect on service evaluation but also influenced consumers' experiences through serial mediation effects involving liberation, hedonic engagement, and fun. These results underscore the critical role of visual representation of VAs in shaping consumer perceptions and reinforce the importance of fun in improving overall service evaluations in online retail environments.

5. Study 2

Study 2 delved deeper into the impact of VA form by examining how variations in form realism—rather than just the presence or absence of form—influence consumers' evaluations of VA services. To achieve this, we employed a one-way between-subjects design (VA form realism: low vs. high), testing H1b, H2b, and H3b.

5.1. Stimuli and procedures

5.1.1. Stimuli development using generative AI: form realism (low vs. high)

We retained the same scenario and video stimulus from the “presence” condition of Study 1 for the “low form realism” condition in Study 2. For the “high form realism” condition, as previously outlined in Section 4.1.1, we utilized a hyper-realistic human image generated by Bing's AI Image Creator during Study 1. This AI-generated “high form realism” VA image was then animated into a video using HeyGen. The final video stimuli for both the low and high form realism conditions in Study 2 are shown in Table I in the Web Appendix.

5.1.2. Pretest: manipulation test

To verify the manipulation, we pretested 110 US participants via Dynata, randomly assigned to low ($n = 54$) or high ($n = 56$) form realism conditions. Following exposure to the respective stimuli, participants were asked to rate their perceptions of the VA's form realism (Lipp et al., 2021; Paauwe, Hoorn, Konijn, & Keyson, 2015) on a 7-point Likert scale ranging from “cartoonish” (1) to “humanlike” (7). An independent samples t -test showed a significant difference in perceived form realism between conditions ($t[108] = -28.2$, $p < .001$). Participants rated the low-form realism VA as more “cartoonish” ($M = 2.20$ [mean ≤ 3]), and the high form realism VA as more “humanlike” ($M = 5.77$ [mean ≥ 5]). These results align with the definitions of low and high form realism by Miao et al. (2022) and Kim et al. (2024), confirming successful manipulation.

5.1.3. Main test: survey design and data collection

Given the successful manipulation, we proceeded with the main experiment using the developed stimuli, employing the same four-phase survey design as in Study 1. In the initial phase, participants were provided with definitions and examples of VAs prior to being shown the video stimuli. In the second phase, participants were randomly assigned to either the low or the high form realism condition. After viewing the video for a minimum of 50 seconds, participants answered a screening question, “Where is the VA in the video?” to ensure they had attentively observed the stimuli. Only those who provided a correct response proceeded to the third phase of the survey. In the third phase, as in Study 1, participants evaluated their assigned VA and the interaction experience, assessing factors such as form realism, liberation, hedonic engagement, fun, and service evaluation. All measures were rated on a 7-point Likert scale ranging from “strongly disagree” (1) to “strongly agree” (7). To maintain participants' continuous attention throughout the questionnaire, we included an additional attention check, asking them to select “somewhat agree” at the center of the scale. Participants who did not select this option were screened out. The measurement items employed

Table 3
Discriminant Validity Assessment Results.

Construct	Study 1				Study 2					Study 3				
	2	3	4	5	1	2	3	4	5	1	2	3	4	5
1. Form realism					.937					.930				
2. Liberation	.932				.565	.918				.738	.904			
3. Hedonic engagement	.836	.902			.717	.835	.918			.788	.830	.899		
4. Fun	.902	.867	.911		.627	.88	.871	.921		.751	.854	.851	.910	
5. Service evaluation	.812	.842	.843	.955	.638	.803	.836	.839	.960	.725	.776	.849	.861	.964

Note. Correlation matrix with square roots of average variances extracted (AVEs) bolded on the diagonal.

Table 4
Hypotheses Test Results with PROCESS Macro Analyses (Model 80).

Hypothesis		Effect	Study 1				Study 2				Study 3			
			Effect	LLCI	ULCI	Result	Effect	LLCI	ULCI	Result	Effect	LLCI	ULCI	Result
H1a	VF→SE	Total	.730	.221	1.236	Significant	—	—	—	—	—	—	—	—
H2a	VF→LB→FUN→SE	Indirect	.187	.059	.356	Significant	—	—	—	—	—	—	—	—
H3a	VF→HE→FUN→SE	Indirect	.095	.044	.195	Significant	—	—	—	—	—	—	—	—
H1b	FR→SE	Total	—	—	—	—	.611	.527	.695	Significant	.702	.631	.774	Significant
H2b	FR→LB→FUN→SE	Indirect	—	—	—	—	.099	.040	.165	Significant	.159	.099	.235	Significant
H3b	FR→HE→FUN→SE	Indirect	—	—	—	—	.103	.041	.170	Significant	.150	.081	.234	Significant

Note. Visual form (VF); form realism (FR); liberation (LB); hedonic engagement (HE); perceived fun (FUN); service evaluation (SE); lower limit of the 95 % confidence interval (LLCI); upper limit of the 95 % confidence interval (ULCI).

in Study 2 are outlined in Table 2. In the final phase of the survey, participants were asked to provide their demographic information.

The survey was administered via Dynata, using the same sampling frame as in Study 1: US residents aged 18 and above. A total of 300 responses were collected, 150 of which were provided by participants in the low form realism condition and 150 by those in the high form realism condition. A power analysis confirmed that the sample size was sufficient, surpassing the minimum requirement of 139 participants for a model with 16 predictors, thus achieving an α value of .05, 80 % power and a medium effect size of .15 (Cohen, 1988). The demographic analysis of the survey participants revealed a balanced gender distribution, with 50 % identifying as female and 50 % as male. The majority of the respondents were Caucasian (68 %), aged either above 55 (26 %) or between 25 and 35 (23 %), 50 % were married, 34 % held a college degree, 54 % were employed full-time, and 39 % reported an annual household income of 25,000–49,999 USD.

5.2. Results

5.2.1. Manipulation check results

An independent samples *t*-test revealed a successful manipulation ($t[298] = -46.33, p < .001$). Specifically, participants in the low form realism condition ($n = 150; M = 1.96$ [mean ≤ 3]) perceived the stimuli as significantly less anthropomorphic than those in the high form realism condition ($n = 150; M = 5.67$ [mean ≥ 5]), thereby confirming successful manipulation.

5.2.2. Common method bias test results

As in Study 1, we employed the marker variable approach to assess potential common method bias in Study 2, following the same procedure. “Attitude toward the color blue” served as the marker variable. A chi-square difference test between the baseline and Method-C models revealed a significant result ($\Delta\chi^2 = 23.71, p < .001$), indicating the presence of method variance. However, further comparisons between Method-C and Method-U ($\Delta\chi^2 = 12.66, p = .63$) and Method-U and Method-R ($\Delta\chi^2 = 0.34, p = 1.00$) showed no significant differences. Therefore, common method bias was not deemed a serious concern in Study 2.

5.2.3. Measurement test results

Since common method bias was not a concern, we proceeded to evaluate the measurement model. CFA results indicated a good model fit: $\chi^2 = 259.84, df = 94, \chi^2/df = 2.76, CFI = .98, TLI = .97, NFI = .97$, and RMSEA = .08 (Gheshlagh et al., 2018). As shown in Table 2, the assessments of construct reliability and convergent validity revealed no issues, with all factor loadings, as well as α , CR, and AVE values exceeding their respective recommended thresholds of .71, .70, .60, and .50, respectively (Cheung et al., 2023; Sarmiento & Costa, 2017; Tabachnick et al., 2013). Additionally, Table 3 confirms discriminant validity using the Fornell-Larcker criterion.

5.2.4. Hypotheses test results

With the manipulation confirmed and the measurement model validated, we proceeded to test the main effect of VA form realism. A one-way ANOVA revealed a significant positive effect of VA form realism (low = 0, high = 1) on VA service evaluation ($F[1, 298] = 136.4, p < .001$), supporting H1b. This effect was further corroborated using the PROCESS macro (Model 80 with 5000 bootstrapped resamples; see Table 4), which showed a significant total effect of form realism on service evaluation (total effect: $\beta = .61, 95\% \text{ CI} = [.53, .70]$). As shown in Table 4, the analysis also confirmed that serial mediation pathways were significant. VA form realism influenced consumers' service evaluation through two sequential mechanisms: one involving liberation followed by fun (indirect effect: $\beta = .10, 95\% \text{ CI} = [.04, .17]$) and the other involving hedonic engagement followed by fun (indirect effect: $\beta = .10, 95\% \text{ CI} = [.04, .17]$). The results indicated that participants interacting with a VA featuring high form realism, compared to low form realism, experienced greater feelings of liberation ($\beta = .57, p < .001$) and hedonic engagement ($\beta = .72, p < .001$). These perceptions, in turn, significantly enhanced their sense of fun—driven by both liberation ($\beta = .51, p < .001$) and hedonic engagement ($\beta = .42, p < .001$)—which ultimately led to more favorable evaluations of the VA service ($\beta = .34, p < .001$). Together, these findings support H2b and H3b.

5.3. Discussion

Building on the foundational insights from Study 1, the findings from Study 2 provide more robust and pronounced evidence that enhancing VA form realism significantly elevates service evaluations. This effect is mediated through the same serial pathways identified in Study 1—via liberation and fun, as well as through hedonic engagement and fun. These results indicate that highly realistic VAs, as employed by contemporary retailers, not only enhance perceived service quality but also do so by intensifying consumers' sense of fun.

6. Study 3

Study 3 aimed to replicate the findings of Study 2 (H1b, H2b, and H3b) using a new consumer sample and to investigate whether online shopping motivation moderates the proposed serial mediation effects (H4). To this end, we employed a 2 (VA form realism: low vs. high) $\times$ 2 (online shopping motivation: task-oriented vs. recreation-oriented) between-subjects design.

6.1. Stimuli and Procedures

6.1.1. Stimuli development: online shopping motivation (task- vs. recreation-oriented)

For the low and high form realism stimuli, we reused the videos developed in Study 2. However, since stimuli for online shopping motivation had not yet been created, Study 3 addressed this by differentiating between task-oriented and recreation-oriented motivations. To that end, we crafted two corresponding textual scenarios based on

methods from Coskun, Gupta, and Burnaz (2019) and Wenzel and Benkenstein (2018), with modifications to better align with our research context. The task-oriented scenario read as follows: “Please imagine you’ve received an invitation to a formal dinner next month. However, you realize that your wardrobe lacks a suitable dress or suit for the occasion. So, you decide to browse online to find and purchase a nice dress or suit that fits your needs. When browsing the website, you click on “Interact with the assistant Clara”. Then, you will see a video where an AI-empowered virtual retail assistant guides you through an online fashion retail store.” In contrast, the recreation-oriented scenario read: “Please imagine a lazy day at home with no particular plans, just feeling bored. You sit in front of the computer, watching videos, yet finding none that catch your interest. Suddenly, an advertisement for an online fashion retail store pops up. Out of curiosity, you decide to click on the store website. When browsing the website, you click on “Interact with the assistant Clara”. Then, you will see a video where an AI-empowered virtual retail assistant guides you through an online fashion retail store.” The stimuli employed for both form realism and shopping motivations are presented in Table I in the Web Appendix.

6.1.2. Pretest: manipulation test

Prior to the main experiment, a pretest was conducted to validate the manipulation of two critical variables: the degree of VA form realism (low vs. high) and consumers’ shopping motivation (task-oriented vs. recreation-oriented). This pretest was administered via Dynata, utilizing a sample of 197 US residents aged 18 and above. Participants were randomly assigned to one of four experimental conditions: (1) a low-form realism VA with task-oriented shopping motivation ($n = 50$), (2) a low-form realism VA with recreation-oriented shopping motivation ($n = 50$), (3) a high-form realism VA with task-oriented shopping motivation ($n = 50$), and (4) a high-form realism VA with recreation-oriented shopping motivation ($n = 47$). The pretest survey was structured into four sections: (1) consent and orientation, (2) scenario presentation, (3) manipulation validation, and (4) demographic information.

To assess the manipulation of VA form realism, participants evaluated the VA’s appearance using a 7-point Likert scale, ranging from “cartoonish” (1) to “humanlike” (7), consistent with the scale used in Study 2. Additionally, to validate the manipulation of shopping motivation, participants responded to a question designed to assess their understanding of the scenario: “Based on the scenario provided, what is your primary reason for visiting the virtual fashion retail store?” Following the methodology proposed by Coskun et al. (2019), we used a two-item scale to measure this aspect: (1) “to acquire some needed products” (task-oriented; reverse-coded) and (2) “to gain enjoyment during the shopping trip and spend leisure time” (recreation-oriented). Participants rated their agreement with each item on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

Independent samples t -test results confirmed the effectiveness of the manipulations for both VA form realism ($t[195] = -10.8, p < .001$) and shopping motivation ($t[195] = -31.5, p < .001$). Specifically, the VA with low form realism ($n = 100$; $M = 2.63$) was perceived as significantly more “cartoonish” compared to the one with high form realism ($n = 97$; $M = 4.71$). While the mean score for the high form realism condition in this pretest was slightly below the operational definition (form realism ≥ 5), this was not deemed problematic, given that the overall mean for this condition, combining pretest results from Study 2 and Study 3 ($n = 153$; $M = 5.10$), exceeded the threshold of five. Nonetheless, to ensure the reliability of the manipulation, we decided to further validate this aspect in the main test. Regarding shopping motivation, the task-oriented group ($n = 100$; $M = 1.95$) had a significantly lower score on the recreational motivation item compared to the recreation-oriented group ($n = 97$; $M = 5.78$).

6.1.3. Main test: survey questionnaire and data collection

After successfully validating all manipulations, we proceeded with the main test. The online survey for Study 3 was structured similarly to

that adopted in Study 2, using the same four-phase survey design. Participants first provided informed consent and reviewed an introduction, which included definitions and examples of VAs in the retail context. They were then randomly assigned to one of four experimental conditions: (1) low-form realism VA with task-oriented shopping motivation, (2) low-form realism VA with recreation-oriented shopping motivation, (3) a high-form realism VA with task-oriented shopping motivation, and (4) high-form realism VA with recreation-oriented shopping motivation. To ensure participants carefully read the scenario and fully understood their assigned shopping motivation, we included a question that served both as an attention check and a manipulation check: “Based on the scenario provided, what is your primary reason for visiting the online fashion retail store?”

Next, as in Study 2, participants viewed the video stimuli for a minimum of 50 seconds to ensure they had ample time to process the content and focus on the VA’s appearance. To confirm their attention, participants were then asked a validation question: “Where is the VA in the video?” Only those who answered this question correctly proceeded to the subsequent manipulation questions. Following this, as in previous studies, participants were asked to rate their experience with the assigned VA, evaluating factors such as form realism, liberation, hedonic engagement, fun, and service evaluation. To maintain participants’ attention throughout the questionnaire, we included an additional attention check, instructing them to select “somewhat agree” at the center of the scale. Participants who did not select this option were screened out. All measures were rated on a 7-point Likert scale, ranging from “strongly disagree” (1) to “strongly agree” (7). The specific measurement items used in the study are detailed in Table 2. The survey concluded with a section eliciting demographic information from participants.

To ensure consistency across all three studies, the survey was administered again through Dynata, targeting US residents aged 18 and older. The final dataset comprised 338 participants, distributed across four conditions: (1) a low-form realism VA with task-oriented shopping motivation ($n = 85$), (2) a low-form realism VA with recreation-oriented shopping motivation ($n = 84$), (3) a high-form realism VA with task-oriented shopping motivation ($n = 84$), and (4) a high-form realism VA with recreation-oriented shopping motivation ($n = 85$). A power analysis confirmed that the sample size exceeded the minimum requirement of 139 participants for a model with 16 predictors, ensuring sufficient statistical power ($\alpha = .05$, 80 % power, with a medium effect size of .15). Most respondents were female (53 %), aged over 55 (30 %), Caucasian (70 %), and married (47 %). Most participants held a bachelor’s degree (35 %), were employed full-time (55 %), and reported an annual household income of 75000–149,999 USD (30 %).

6.2. Results

6.2.1. Manipulation check results

Independent samples t -tests confirmed successful manipulations of both form realism ($t[336] = -50.89, p < .001$) and online shopping motivation ($t[398] = -19.86, p < .001$). Low-form realism stimuli ($n = 169$; $M = 2.02$ [mean ≤ 3]) were perceived as significantly more “cartoon-like” than high realism stimuli ($n = 169$; $M = 5.70$ [mean ≥ 5]). The task-oriented scenario ($n = 169$; $M = 2.70$) was seen as more “purposeful” than the recreation-oriented one ($n = 169$; $M = 4.78$).

6.2.2. Common method bias test results

Using the same marker variable approach as in studies 1 and 2 (“attitude toward the color blue”), Study 3 assessed common method bias. While a significant difference between the baseline and Method-C models ($\Delta\chi^2 = 15.82, p < .001$) indicated some method variance, non-significant differences between Method-C and Method-U ($\Delta\chi^2 = 23.56, p = .07$) and between Method-U and Method-R ($\Delta\chi^2 = 0.35, p = 1.00$) suggest that common method bias is not a serious concern in Study 3.

6.2.3. Measurement test results

CFA results indicated excellent model fit: $\chi^2 = 227.25$, $df = 94$, $\chi^2/df = 2.42$, CFI = .98, TLI = .98, NFI = .97, and RMSEA = .06. Tests of convergent and discriminant validity further confirmed the robustness of the constructs. All factor loadings surpassed the threshold value of .71, indicating excellent construct reliability (see Table 2; Tabachnick et al., 2013). In addition, the α , CR, and AVE values exceeded the recommended thresholds of .70, .60, and .50, respectively, supporting strong convergent validity. As shown in Table 3, the instrument also exhibited discriminant validity (Cheung et al., 2023).

6.2.4. Robustness test results

Since the measurement model was robust, we reexamined H1b, H2b, and H3b to validate the robustness of Study 2's findings. Using one-way ANOVA and the PROCESS macro (Model 80 with 5000 bootstrapped resamples), the results replicated those of Study 2. The main effect of VA form realism on consumer service evaluation was significant in both ANOVA ($F[1, 336] = 262.6$, $p < .001$) and PROCESS (total effect: $\beta = .70$, 95 % CI = [.63, .77]), supporting H1b. As shown in Table 4, the two serial mediation effects were also significant. High (versus low) VA form realism had stronger indirect effects on service evaluation via liberation and fun (indirect effect: $\beta = .16$, 95 % CI = [.10, .24]) and hedonic engagement and fun (indirect effect: $\beta = .15$, 95 % CI = [.08, .23]). Specifically, high form realism significantly increased perceptions of liberation ($\beta = .74$, $p < .001$) and hedonic engagement ($\beta = .79$, $p < .001$), which in turn enhanced fun ($\beta = .44$ and $\beta = .39$, respectively, $p < .001$), ultimately boosting service evaluations ($\beta = .50$, $p < .001$).

6.2.5. Moderated mediation test results

After confirming the robustness of Study 2 findings, we examined the moderating effect of consumers' online shopping motivation on the serial mediation models using the PROCESS macro (Model 83 with 5000 bootstrapped resamples). The analyses revealed significant interaction effects of form realism (measured on a 7-point scale) and shopping motivation (task-oriented = 0, recreation-oriented = 1) on liberation ($\beta = .23$, $p < .001$) and hedonic engagement ($\beta = .12$, $p < .05$), as shown in Fig. 2. To further validate these results, we conducted two-way ANOVA tests (see Table J and Figure K in the Web Appendix), which reconfirmed the interaction effects.

As shown in Table 5, the PROCESS macro (Model 83) analysis results indicate that shopping motivation significantly moderates the "VA Form Realism $\rightarrow$ Liberation $\rightarrow$ Fun $\rightarrow$ VA Service Evaluation" serial mediation pathway (moderated mediation index: $\beta = .11$, 95 % CI = [.04, .19]). Specifically, the positive serial mediation effect was stronger for consumers with a recreation-oriented shopping motivation (indirect effect: $\beta = .36$, 95 % CI = [.24, .50]) compared to those with a task-oriented motivation (indirect effect: $\beta = .25$, 95 % CI = [.17, .35]). Similarly, shopping motivation also moderated the "VA Form Realism $\rightarrow$ Hedonic Engagement $\rightarrow$ Fun $\rightarrow$ VA Service Evaluation" serial mediation pathway (moderated mediation index: $\beta = .04$, 95 % CI = [.00, .10]). Once again, this serial mediation effect was more pronounced for consumers with a

recreation-oriented shopping motivation (indirect effect: $\beta = .28$, 95 % CI = [.18, .39]) than for those with a task-oriented motivation (indirect effect: $\beta = .23$, 95 % CI = [.15, .33]), supporting H4.

6.3. Discussion

The findings from Study 3 underscore the robustness of the positive main effect of VA form realism on service evaluations, as well as the serial mediation effects through liberation, hedonic engagement, and fun. More importantly, the results provide additional insights into how consumer motivations influence these relationships. As anticipated, the serial mediation effect was stronger for consumers with a recreational online shopping motivation compared to those with a task-oriented motivation. This distinction highlights the essential need to tailor VAs' anthropomorphic features and the overall fun experience to better match consumers' motivations for online shopping.

7. Discussion

Advancements in generative AI have significantly enhanced the capabilities of VAs, enabling them to possess visual embodiments that are increasingly sophisticated and human-like. As a result, retailers are now adopting VAs with distinct forms and even high levels of visual realism, moving away from the formless VAs commonly used in the past. However, despite these industry advancements, academic research has largely treated VAs as formless entities, paying limited attention to the effects of VA form realism compared to the more extensively studied cognitive and behavioral realism. Not only is research on VA form realism relatively scarce, but the mixed findings in the existing literature—specifically regarding whether greater visual realism elicits positive or negative consumer responses—highlight the need for a comprehensive reassessment. To advance the field, it is also essential to address the theoretical limitations and conceptual gaps in prior studies. Many VA studies have relied on traditional frameworks rooted in technology-, social-, or cognitive-oriented paradigms that primarily view VAs as technological tools, social agents, or cognitive aids. While these perspectives provide valuable insights, they often overlook the emotionally nuanced and experientially rich nature of consumer–VA interactions in online retail. Our study addresses these gaps by offering new theoretical insights into the relationship between VA form, realism, and consumer behavior through the lens of consumer fun theory. By focusing on the mediating roles of liberation, hedonic engagement, and fun, we more accurately capture the underlying affective and psychological mechanisms driving consumer responses. These findings deepen our theoretical understanding of how VA design influences consumer behavior and offer valuable guidance for improving the development and deployment of VAs in online retail environments.

7.1. Theoretical implications

First, our findings reveal that the mere presence of a VA with a

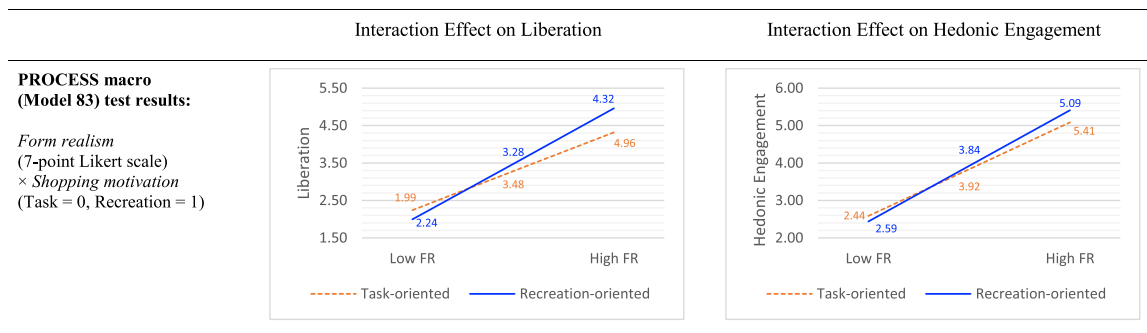


Fig. 2. Interaction Effects of Form Realism and Online Shopping Motivation on Liberation and Hedonic Engagement.

Table 5
Moderated Mediation Analysis Results with PROCESS Macro Analyses (Model 83).

Hypothesis		Type	Moderator	Conditional sequential mediation effects						Index of Moderated Mediation	LLCI	ULCI	Result
				Task-oriented condition (MV = 0)			Recreation-oriented condition (MV = 1)						
				Effect	LLCI	ULCI	Effect	LLCI	ULCI				
H4	FR×MV→LB→FUN→SE	Indirect	MV	.253	.169	.347	.361	.244	.495	.108	.043	.191	Significant
	FR×MV→HE→FUN→SE			.233	.153	.327	.277	.179	.389	.044	.003	.095	Significant

Note. Form realism (FR); liberation (LB); hedonic engagement (HE); perceived fun (FUN); service evaluation (SE); online shopping motivation (MV, 0 = task-oriented, 1 = recreation-oriented); lower limit of the 95 % confidence interval (LLCI); upper limit of the 95 % confidence interval (ULCI).

visible, recognizable form—compared to the absence of any visual representation—significantly enhances consumer evaluations of the VA service. Furthermore, VAs with higher levels of human-like form realism—compared to those with less lifelike appearances—further enhance positive consumer evaluations, as evidenced by a stronger intention to reuse the service. These insights are especially meaningful because, despite prior research producing inconclusive results regarding the impact of VA form realism, our study provides robust empirical evidence that both the presence of a perceptible visual form and the degree of its form realism substantially improve consumer responses. In particular, our findings reinforce earlier scholarship that emphasizes the superiority of form-based VAs over formless, behavior-driven chatbots (Go & Sundar, 2019; Qiu & Benbasat, 2009). Furthermore, our findings support prior arguments that VA form realism enhances consumer perceptions (Qiu & Benbasat, 2009; Song et al., 2024), while contradicting studies that raised concerns about consumer discomfort when VAs appear too human-like (Pentina et al., 2023; Thaler et al., 2020).

These findings carry important theoretical implications in two key ways. First, our study addresses a longstanding contextual bias in VA research, which has disproportionately focused on formless or minimally embodied agents, despite growing industry adoption of visually rich, photorealistic VAs. By aligning empirical inquiry with real-world design trends—particularly in experience-driven sectors like fashion retail—we validate the positive impact of highly realistic, human-like VAs on consumer responses. By doing so, we move beyond the previously narrow focus on cognitive and behavioral realism, foregrounding the critical role of form realism as a key driver of user experience in digitally mediated service environments. Second, our findings build on existing research that emphasizes the importance of visual interface design in online retail, which has traditionally focused on elements such as website layout (Ramezani Nia & Shokouhyar, 2020), color schemes (Koo & Park, 2017), and product display strategies (González, Meyer, & Toldos, 2021). We argue that the form realism of VAs should be recognized as an equally vital component of visual interface design in the context of online retail. This complements and extends prior research, which has primarily emphasized the effects of VAs' cognitive realism (Rust & Huang, 2021; Schepers et al., 2022) and behavioral realism (Li & Wang, 2023) on consumer perceptions and VA performance. Our study clearly demonstrates that VA research must evolve by repositioning form realism from a peripheral concern to a central factor that significantly shapes consumer attitudes, engagement, and overall experience.

Beyond the main effects of VA form and realism on service performance, our study uncovers nuanced psychological mechanisms that explain these effects by building on consumer fun theory. Specifically, we found that consumers perceive VAs with a visible form—and those exhibiting higher human-like appearance realism—as delivering a superior service compared to their less lifelike or formless counterparts. This is because such VAs evoke a stronger sense of liberation, greater hedonic engagement, and, ultimately, heightened feelings of fun among consumers during their interactions with the VA. These experiential and emotional responses are pivotal in shaping consumers' service evaluations and attitudes toward the VA: (1) VA form and realism enhance consumers' sense of liberation, which increases feelings of fun,

ultimately leading to more positive evaluations; and (2) they heighten hedonic engagement, which likewise fosters greater fun and results in more favorable evaluations.

These findings carry important theoretical implications. First, they correct a longstanding theoretical bias prevalent in VA literature. Historically, researchers have primarily treated VAs as technical tools, employing technology-oriented theories that emphasize efficiency and functionality (Chen et al., 2021; Nayak & Nair, 2025). Alternatively, some studies have applied social-oriented theories that highlight the interpersonal capabilities of VAs (Cheng et al., 2024; Song et al., 2022). Others have adopted cognitive-oriented theories that focus on how consumers process information from VAs and how this affects cognitive evaluations, such as decision-making or task performance (Weiler et al., 2022; Yang et al., 2025). While these approaches offer valuable insights, they often neglect the hedonic and experiential dimensions of VA interactions—particularly in leisure-driven contexts like fashion retail. Our research is novel in expanding the scope of VA literature to incorporate hedonic and emotional motivations in consumer interactions, emphasizing the role of fun as a central driver of consumer evaluations. By doing so, we bridge an important gap and offer a more comprehensive understanding of how VAs influence consumer experiences in non-utilitarian settings. Second, these findings provide additional theoretical contributions by offering empirical support for the application of consumer fun theory in the VA context. Specifically, our research demonstrates that VAs with a visible form (versus formless VAs) and higher form realism (versus lower realism) play a critical role in enhancing consumers' interactions by fostering fun, which is derived from feelings of liberation and hedonic engagement. This finding is important because it underscores the modern consumer's growing demand for experiences that are liberating, engaging, and emotionally fulfilling. It highlights the need for VA research to prioritize designing VAs not only as functional tools or social agents but also as enablers of enjoyable experiences. By doing so, researchers and practitioners can better align VA design with the evolving preferences of consumers, particularly in hedonic and experiential retail contexts.

Last, our research provides nuanced theoretical insights into the VA literature by exploring the dynamic interplay between VA realism and consumers' shopping motivations—task-oriented versus recreation-oriented—in shaping service evaluations. Our analyses offer compelling evidence that the serial mediation effects identified in this study are significantly stronger for consumers driven by recreational shopping motivations relative to their task-oriented counterparts. Specifically, for recreational shoppers, the realism of a VA's form triggers greater emotional liberation, enhances hedonic engagement, and amplifies fun, leading to more favorable service evaluations. In contrast, for task-oriented consumers, these effects are less pronounced. These results highlight the importance of contextualizing VA research within the unique dynamics of online retail, where consumers often seek not just products but also engaging and enjoyable experiences. Unlike physical stores that prioritize functionality, online environments frequently serve leisure-oriented goals such as escapism and emotional gratification. Accordingly, building on these findings, VA interface designs should be strategically tailored to align with shoppers' underlying motivations—enhancing effectiveness by adapting form realism to suit various

shopping contexts and consumer profiles.

7.2. Practical implications

Our findings provide significant insights and actionable recommendations for retailers and VA interface designers. First, our study challenges the prevailing assumption that highly human-like VAs may induce discomfort or unease. In contrast to these concerns, we demonstrate that advanced visual enhancements in VAs substantially elevate consumer perceptions, transforming interactions into more engaging and more enjoyable experiences. Building upon these insights, we encourage retailers to confidently adopt AI-generated VAs with visual representations and anthropomorphic features. Specifically, our research indicates that VAs with a visual form are far more effective in eliciting positive consumer evaluations compared to traditional voice-based assistants, which often lack the immersive qualities that contemporary consumers increasingly seek. Furthermore, when comparing VAs with various visual characteristics, those with highly human-like appearances were significantly more effective in enhancing consumer satisfaction than their cartoonish or animated counterparts. In light of these findings, we recommend that retailers prioritize the integration of visually realistic features in their VA interfaces. Such sophisticated VAs will not only enhance consumer engagement with the VA services but also enrich the overall online shopping experience, making it more immersive, enjoyable, and captivating.

Second, our findings provide valuable guidance for VA interface designers, particularly in enhancing consumers' enjoyment of the technology. To foster a sense of liberation, hedonic engagement, and ultimately, consumer fun, we recommend that VA interfaces incorporate features that promote freedom and immersion, allowing consumers to experience escapism and enjoyment during their interactions. One effective way to promote this sense of liberation is for designers to adopt informal, approachable visual styles or attire for VAs, rather than rigid, formal designs. This approach can help create a more relaxed atmosphere and enable users to disconnect from the stresses associated with work-related pressures or other obligations. Additionally, we recommend tailoring the VA's appearance to enhance hedonic engagement, drawing consumers into a more pleasurable and immersive experience. Incorporating features, such as natural skin tones, age-appropriate traits, and regionally relevant physical characteristics, can help users feel more connected and "in the moment" with the VA, fostering an environment conducive to enjoyment. As our study suggests, this sense of enjoyment can significantly enhance the overall interaction. By integrating these elements, interface designers can create VAs that not only meet consumers' functional needs but also enrich their shopping experience, making it more enjoyable and leading to more positive service evaluations.

Last, it is crucial for retailers and VA designers to account for consumers' underlying shopping motivations, as these significantly influence the impact of VA form realism on service evaluations. Our findings indicate that VAs with high form realism are particularly effective in fostering a sense of liberation among recreationally oriented consumers, in contrast to task-oriented shoppers. In light of this distinction, online retailers should analyze consumer behavior to identify periods when recreational shoppers are most active. For example, retailers can track specific times, dates, or shopping events—such as weekends, holidays, or promotional periods—when recreational shoppers tend to engage more frequently. During these peak intervals, deploying VAs with life-like features can enhance consumers' sense of escapism and hedonic engagement, ultimately promoting greater enjoyment and prolonged interactions. Conversely, during periods dominated by task-oriented shoppers, the emphasis on high form realism may be less critical, as these consumers prioritize practicality over immersive experiences.

7.3. Limitations and future research

While this study offers valuable theoretical and practical contributions, it also has certain limitations that open up meaningful avenues for future research. First, we employed video-based stimuli to manipulate and assess the effects of VA form and form realism. As discussed in [Section 4.1.1](#), the video treatment was appropriate for our objectives, offering high control and consistent exposure across conditions. However, future researchers may note that while video stimuli are effective for standardized manipulation, they inherently limit participant interaction by providing a static, passive viewing experience rather than dynamic, real-time engagement. To address this limitation, future studies could incorporate more interactive and immersive formats—such as mock applications developed using tools like ProtoPie or Figma. ProtoPie enables the creation of highly interactive prototypes that include animation and responsive behaviors, effectively simulating real-world mobile or web shopping environments. Unlike video, these prototypes allow participants to engage in dynamic interactions with the VA (e.g., selecting products), thereby enhancing ecological validity and producing deeper insights into user experience. Figma, meanwhile, supports the design of clickable interfaces. These interactive treatments would offer a more lifelike context for studying how VA form and realism shapes consumer perceptions and behavior. Second, our study focused exclusively on the online fashion retail context and used consumer fun theory to explore consumers' hedonic and experiential responses to VAs. While fashion is a representative hedonic category, future research should investigate whether our findings generalize to other hedonic online sectors such as beauty and cosmetics, luxury goods, or home décor. Third, our study did not explicitly control for several potential confounding variables that may influence how consumers evaluate VAs. These include VAs' perceived physical attractiveness and consumers' prior familiarity with VA technology. Prior research suggests that attractiveness can bias user evaluations, while familiarity may moderate emotional responses—either amplifying or diminishing perceived novelty or engagement. Future research should consider incorporating these variables as controls or moderators to better understand how individual differences interact with VA design features in shaping service perceptions. In sum, future work should explore more interactive experimental designs, broader retail contexts, and individual-level moderators. Such efforts will strengthen both the theoretical robustness and practical relevance of research on VA design, guiding the development of emotionally engaging, consumer-centered AI technologies in online commerce.

8. Conclusion

This study addresses a significant gap in the existing literature by investigating how VA form and realism—further enhanced by the latest generative AI technologies—affect consumers' evaluations of VA services, particularly through the newly identified factor of fun. Our analyses reveal two main effects, showing that introduction of anthropomorphic visual interface VA (compared to no visual form) and higher perceptibility of form realism in a VA (compared to lower realism) improve consumer evaluations of the VA service. More importantly, we uncovered the psychological mechanisms driving this effect, identifying two distinct serial mediation pathways: (1) VA form and realism → enhance liberation → increase fun → more favorable evaluations of the VA service; and (2) VA form and realism → boost hedonic engagement → increase fun → more favorable evaluations of the VA service. Additionally, our findings suggest that consumers' shopping motivations—whether recreation-oriented or task-oriented—moderate these mechanisms.

Collectively, these findings make three key contributions to the VA literature. First, they address persistent contextual bias, where prior studies have disproportionately focused on formless VAs—despite growing industry adoption of visually embodied agents. Second, they

highlight the importance of form realism, a dimension underexplored relative to behavioral or cognitive realism. Third, they challenge the dominance of technology-, social-, and cognitive-oriented theories, which tend to overlook hedonic and affective processes. By applying consumer fun theory, we introduce an emotional and experiential lens that is better suited to understanding VA interactions in hedonic retail settings, such as online fashion retail. Beyond theoretical implications, our findings offer actionable guidance for VA designers and online retailers. They underscore the need to move beyond a purely functional or social lens and recognize the value of designing VAs that evoke liberation, hedonic engagement, and fun—elements increasingly central to the online shopping experience. By doing so, our study shows how AI-driven VAs can not only improve service effectiveness but also make digital retail more enjoyable, immersive, and emotionally satisfying for consumers.

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CRediT authorship contribution statement

Hanfei XUE: Conceptualization, Formal analysis, Methodology, Writing – original draft. **Jiayu WANG:** Methodology, Writing – original draft. **Szeman CHONG:** Data curation, Formal analysis, Methodology, Validation, Writing – original draft, Visualization. **Woojin CHOI:** Data curation, Formal analysis, Validation, Writing – original draft. **Chung-Wha (Chloe) KI:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review and editing. **Christina W. Y. WONG:** Supervision, Writing – review and editing.

Declaration of Competing Interest

This research was supported by both an industry donation and funding from the authors' academic institution. However, the authors declare that they have no financial conflicts of interest or personal relationships that could have influenced the conduct or outcomes of this research.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ijinfomgt.2025.102984](https://doi.org/10.1016/j.ijinfomgt.2025.102984).

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